

Number Play — Full Test

Mixed questions from all three concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes · Total: 28 marks. Answers on the separate key.

Objective (1 mark each)

Q1 · MCQ**[1]**

How many different expressions can you form by placing '+' or '-' signs between four consecutive numbers?

- (a) 4 (b) 6 (c) 8 (d) 16

Q2 · MCQ**[1]**

Which expression is always even for every integer n ?

- (a) $2n + 1$ (b) $3n$ (c) $4n$ (d) $n + 1$

Q3 · MCQ**[1]**

A number is divisible by 9 if:

- (a) it ends in 9 (b) the sum of its digits is divisible by 9 (c) it ends in 0 (d) it is even

Q4 · MCQ**[1]**

How many multiples of 7 are there between 200 and 300?

- (a) 13 (b) 14 (c) 11 (d) 15

Q5 · FILL IN THE BLANK**[1]**

The digital root of 7309 is _____.

Q6 · TRUE / FALSE

[1]

$n(n + 1)$ is always even for any integer n .

True False

Q7 · MCQ

[1]

The number 405 is divisible by:

(a) 2 (b) 4 (c) 9 (d) 11

Q8 · ASSERTION–REASON

[1]

Assertion (A): 1 is a prime number.

Reason (R): Prime numbers have exactly two factors.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

The greatest of seven consecutive integers is k . Write the other six numbers in terms of k .

Q10

[2]

Find the smallest and greatest 4-digit number divisible by 9.

Q11

[2]

Find all values of z so that $31z5$ is a multiple of 9.

Q12

[2]

Show, using algebra, that $2a + 2b$ is always even.

Long answer (3 marks each)

Q13

[3]

The sum of four consecutive numbers is 34. Find the numbers, showing your method.

Q14

[3]

Determine whether 406839 is divisible by 2, 3, 9 and 11, using the divisibility tests.

Q15

[3]

Solve the cryptarithm $AB \times A = CCC$. Find A, B and C.

Q16

[3]

Find the digital root of $63p + 18q + 29$ for any integers p and q , and explain why it is fixed.

Total: 28 marks · Check your work against the **answer key**.